

Solving Differential Equations with Series

MATH 146 – Calculus II

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Example: First Order Linear Equations

Example: Evaluate $y' - y = 0$.

Example: First Order Linear Equations

Example: Use power series to solve $y'' + x^2y = 0$ with the initial condition $y(0) = a$ and $y'(0) = b$

Example: First Order Linear Equations

Example: Suppose that $\sum_{n=0}^{\infty} a_n x^n$ converges to a function y such that $y'' - y' + y = 0$ where $y(0) = 1$ and $y'(0) = 0$.

Find a formula that relates a_{n+2} , a_{n+1} and a_n and compute $a_0, \dots, a_5$.